Design and Analysis of a PLL-Based Closed-Loop Buck/Boost DC–DC Converter with Arduino PWM Control

A Stage-by-Stage Mathematical Framework with Hardware Implementation for Regulated 10 V $\longrightarrow$ 5 V Conversion

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Aim and Objective. To design and implement a PLL-assisted adaptive PWM control scheme for a Buck/Boost converter capable of maintaining a regulated output voltage by dynamically adjusting the duty cycle through phase synchronization between the output ripple feedback and an externally generated reference frequency.

Background and Motivation. Efficient and regulated DC–DC power conversion is a foundational requirement in embedded systems, battery-powered devices, and laboratory power supplies. Conventional open-loop buck converters suffer from output-voltage drift under varying load and line conditions, motivating the need for a robust closed-loop architecture. This work presents the complete design, hardware implementation, and mathematical analysis of a *Phase-Locked Loop (PLL)-based buck/boost converter* that regulates an input supply of $V_{in} = 10\,\mathrm{V}$ to a stable output of $V_{out} = 5\,\mathrm{V}$, delivering a 50% step-down with tight voltage regulation across load transients.

System Architecture. The proposed system integrates six functional stages into a coherent closed-loop topology. The power stage is an N-channel MOSFET-switched buck (or boost) converter driven by a PWM gate signal of duty cycle $D = V_{out}/V_{in}$. The output is sensed through a resistive divider (R_1–R_2), scaled to a feedback voltage $V_{fb} = V_{out} \cdot R_2/(R_1 + R_2)$. An AC coupling capacitor extracts the switching-ripple component and feeds it to the phase detector, which compares it against the Arduino-generated reference f_{ref}. The phase-error voltage $V_{PD} = K_{PD}\,\phi_{err}$ drives the VCO, whose output $f_{VCO} = f_0 + K_{VCO}\,V_{ctrl}$ is compared with a sawtooth ramp inside a comparator, producing the PWM signal that closes the loop.

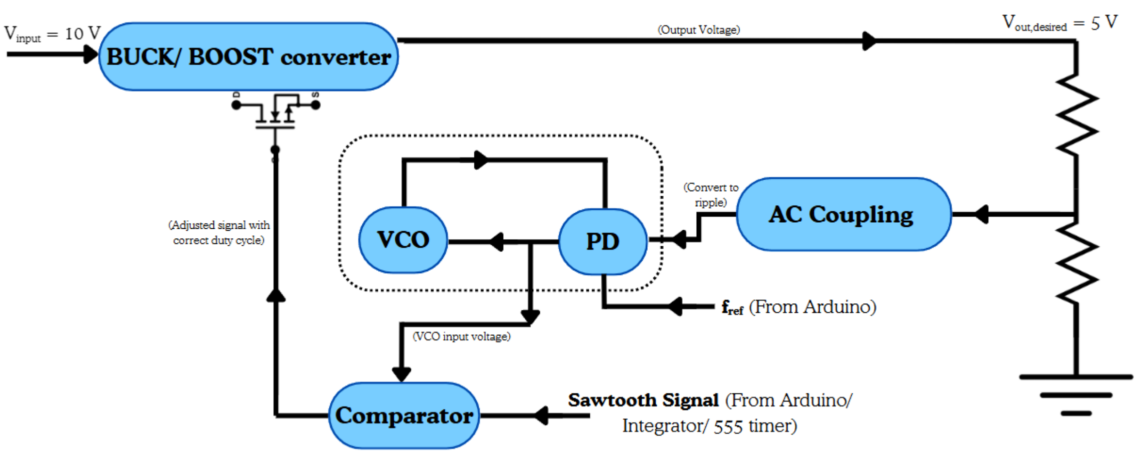


Figure: Complete closed-loop control block diagram of the PLL based Buck/ Boost Converter to hold the output steady .

Figure 1. System Architecture Block Diagram

Work Completed. The following tasks have been completed and are documented with full symbolic derivations and hardware measurements.

(i) PLL Hardware Implementation. The PLL has been fully constructed in hardware. The phase detector is implemented using a **BJT-based differential mixer circuit built around the BC547A** NPN transistor, whose output is buffered before being fed to the next stage. The VCO is designed around a **555-timer-inspired topology**, producing a variable output frequency of up to approximately **500** Hz in response to the PD control voltage. The hardware PD characteristic (output voltage vs. phase difference in degrees) has been measured and curve-fitted using Desmos, yielding the sinusoidal regression:

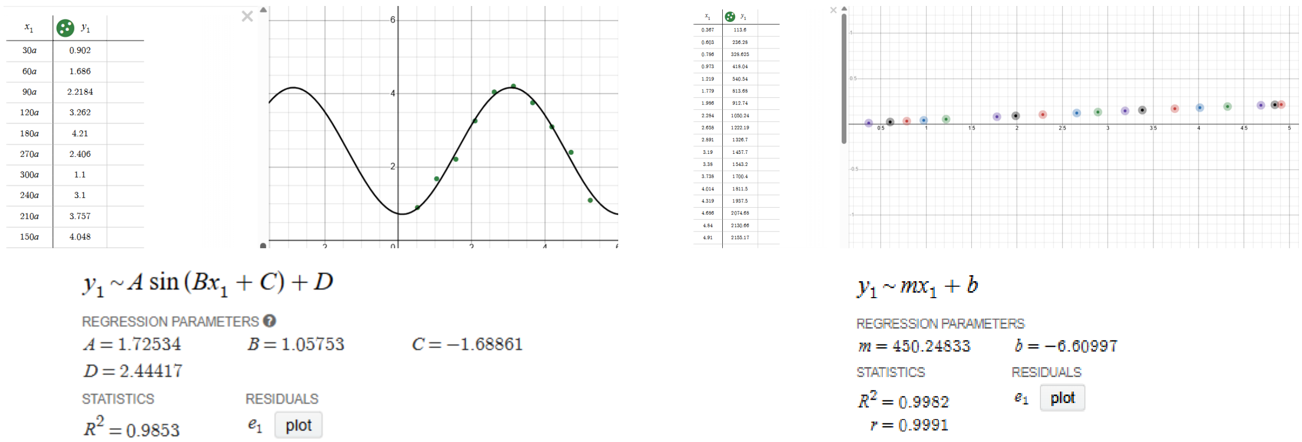
$$V_{PD}(\phi) = A\sin(B\phi + C) + D, \quad A = 1.725,\ B = 1.058,\ C = -1.689,\ D = 2.444, \quad R^2 = 0.985$$

The VCO frequency-vs-voltage characteristic has been measured and confirms excellent linearity, with the linear regression $f = m\,V_{ctrl} + b$ giving $m = 450.25\,\text{Hz/V}$ and $b = -6.61$, $R^2 = 0.998$ — establishing the hardware VCO gain as $K_{VCO} = 450.25\,\text{Hz/V}$.

(ii) Closed-Loop PLL Analysis and Compensator Design. The complete open-loop transfer function $G_{OL}(s) = K_{PD}\,K_{VCO}\,F(s)/s$ has been formulated and analysed using frequency response methods (Bode plots). Natural frequency $\omega_n = \sqrt{K_{PD}\,K_{VCO}/(R_{LF}\,C_{LF})}$ and damping ratio $\zeta = \frac{1}{2}\sqrt{K_{PD}\,K_{VCO}\,R_{LF}/C_{LF}}$ were derived and tuned to achieve optimal lock range, pull-in range, and lock time. A **compensator** has been designed for the loop to meet the required phase margin ($\geq$45) and loop bandwidth ($f_c \approx f_{sw}/10$), ensuring stable and fast acquisition of lock.

(iii) Buck Converter Design. The basic buck converter stage has been designed to output $V_{out} = 5\,\text{V}$ from $V_{in} = 10\,\text{V}$ at $D = 0.5$, with appropriate component sizing and a switching frequency of $f_{sw} = 5\,\text{kHz}$ via Arduino Timer 1 (ICR1 = 511, OCR1A = 256). The feedback mechanism connecting the PLL output to the MOSFET gate driver — completing the full closed-loop regulation — **is not yet implemented** and constitutes the primary remaining deliverable.

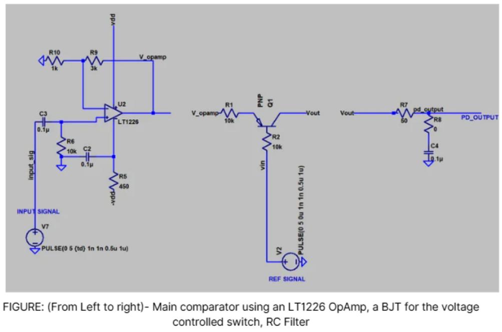
Experimental Results. Figures 2a and 2b show the hardware characterisation data measured from the built circuit and plotted in Desmos. Figure 2a presents the PD output voltage as a function of input phase difference (in units of radians), with the sinusoidal regression confirming the theoretical $K_{PD} \approx V_{DD}/2\pi = 1.902\,\text{V/rad}$ in theory and matching the fitted amplitude $A = 1.725\,\text{V}$. Figure 2b presents the VCO output frequency (Hz) vs. control voltage (V), demonstrating the highly linear $K_{VCO} = 450.25\,\text{Hz/V}$ characteristic required for predictable PLL behaviour.



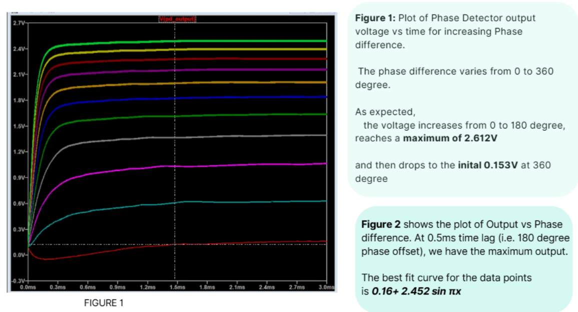
(a) PD hardware output: V_{PD} vs. phase difference. Fitted curve yields $R^2 = 0.985$, $K_{PD} = 1.7253\,\text{V/rad}$ via sinusoidal fit.

(b) VCO hardware output: Frequency vs. V_{ctrl}. Fitted curve yields $K_{VCO} = 450.25\,\text{Hz/V}$, $f = 450.25V - 6.61$, $R^2 = 0.998$ via linear fit.

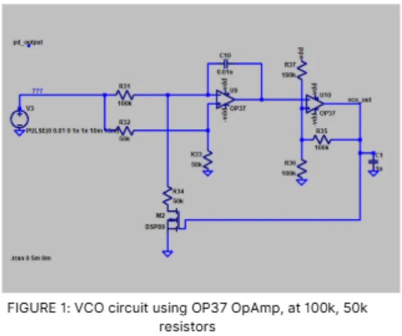
Figure 2. *Hardware characterisation plots obtained from the built BC547A phase detector and 555-timer VCO circuit. All regressions were performed in Desmos.*



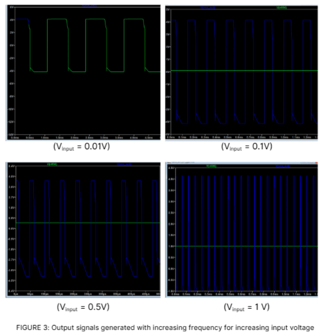
(a) Phase detector schematic circuit using Bipolar Junction Transistor.



(b) Phase detector simulated output voltage response vs. increasing phase difference.



(a) VCO schematic model and verification using an OP37 OpAmp configuration.



(b) Increasing output frequencies with increasing input voltage.

Figure 4. *LTSpice characterisation plots obtained from the simulated phase detector and VCO circuits.*

What This Work Will Demonstrate. The remaining work will close the loop by connecting the PLL output to the buck converter switching MOSFET and demonstrating full closed-loop voltage regulation. Specifically, the following results are targeted: (a) stable $V_{out} = 5.0 \pm 0.1\,\text{V}$ from $10\,\text{V}$ across varying load; (b) oscilloscope verification of $D = 50\%$ PWM at the MOSFET gate; (c) PLL lock acquisition waveforms confirming phase detector and VCO hardware operation at the predicted K_{PD} and K_{VCO} values; (d) a completely locked output voltage inspite of fluctuations in the input voltage.

This abstract accompanies the full technical report containing the system block diagram, complete design summary, key waveform plots, compensator design progress, and annotated Arduino firmware. All numerical values correspond to $V_{in} = 10\ V$, $V_{out} = 5\ V$, $I_{out} \leq 1\ A$, $f_{sw} = 5\,kHz$.